Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 258 (2025) 67–75

Procedia Computer Science

www.elsevier.com/locate/procedia

International Conference on Machine Learning and Data Engineering

Virtual Assistance System for Visually Impaired people using Deep Learning

Faiyaz Ahmad[a], Mohd Zeeshan Ansari[a*], Mohd Tayyab[a], Sharique Shuja[b], Muhammad Zaki[b]

[a]*Department of Computer Engineering, Jamia Millia Islamia, New Delhi, India*
[b]*Department of Electronics & Communication Engineering, Jamia Millia Islamia, New Delhi, India*

Abstract

Assistance is an important necessity for visually impaired people, as it helps them to navigate in challenging conditions in day-to-day life. Object detection, distance and position estimation of nearby objects are crucial tasks to user navigation, which helps to navigate them in outdoor environments. Consequently, utilizing the knowledge of the surroundings by the caption generation of view is also essential to assistance. This paper presents a novel framework that includes object detection, distance and position estimation to generate a generic description of the view. The framework uses best in class pre-trained Yolov5 for object detection, whose inference speed is better than its predecessor, subsequently, a custom model trained on a Kitty dataset for distance estimation. Attention-based model for caption generation, which uses Efficient-Net architecture as the backbone for feature extraction. It is observed that the implemented distance model provides best results in 3 out of 4 parameters, while the proposed Caption Generation framework provides best BLEU scores among multiple models explored. Yolov5 provides 42.5 mAP on COCO dataset. For the Kitty dataset Custom distance estimation model with 296000 parameters, a test RMS Log of .0232 and .1627 squared relative error is obtained.

© 2025 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the International Conference on Machine Learning and Data Engineering

Keywords: Deep Learning, Visual Assistant System, Image Captioning, Efficient Net, Object Detection, Distance Estimation;

1. Introduction

Millions of individuals throughout the world are now impacted by vision impairment. The World Health Organization estimates that 253 million individuals worldwide suffer from vision impairment. The field of study in assistive computer vision for real-time path guidance is under extensive research [1]. The outdoors is one of the primary settings where technology may aid those who are blind [2]. By expanding the area that they can perceive around them and by

1877-0509 © 2025 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the International Conference on Machine Learning and Data Engineering
10.1016/j.procs.2025.04.196

providing precise descriptions of the surroundings, technology might increase their perception power of outdoor surroundings. The goal of the study system was to develop a virtual aid system that will make outdoor tasks easier for anyone who is sight impaired. This system consists of three parts: EfficientNet-B7 with LSTM-based encoder-decoder for picture captioning, YOLOv5 for object detection, and a proprietary neural network trained on the KITTI dataset for position and distance estimates. The basic concept is to use a wearable or smartphone to process images from a camera server and identify objects, distances, and orientations. An image captioner describes what's around the user. Currently, our system provides more generic descriptions and helps to have a broader description of surroundings. The surroundings are majorly urban settings like that of highways. This data could be gathered, and auditory feedback could be sent back to the user's device. This paper is organized as follows: Sec. 2 provides the related work. The subsequent Sec. 3 explain the methods used in the virtual assistance system. Sec. 4 presents the architecture of the proposed work. Sec. 5 carries out experimental results and discussion. Sec. 6 includes the conclusion and future directions.

2. Related Work

A system is proposed in [3] to improve visual perception in visually impaired people. The device uses advanced machine learning techniques to create a three-dimensional environment with recognized boundaries using wearable camera data. Smartwatches have haptic feedback, and the Depth Estimator now brings impediments closer. The 2D Object Detector properly identified city vehicles like cars and buses. In [4], the authors produce accurate depth maps from a single RGB image. Their CNN architecture uses an encoder-decoder structure and a transformer-based block called AdaBins. This block dynamically determines depth bin centers, improving depth map production.CL-HRA, a new image captioning method [5], uses group learning and hierarchical refined attention (HRA) to produce high-quality captions. This approach optimizes captions by emphasizing semantic and visual factors. The study [4] describes an indoor airplane detection system. Wearable technology helps visually challenged people navigate safely by identifying passable places. The device computes plane lengths and supplements a white cane for independent mobility. The navigation system [6] uses a smartphone and server to detect obstacles via speech notifications. Optimizing for environmental characteristics yields 60% recognition. Finally, [7] suggests using a wearable assistive device to help visually impaired people cross streets safely. To detect obstacles, ZED-2 depth maps are merged using a Fast-SCNN model and Yolov5. It’s effective, but users should use a white cane for safety.

Recent research shows AI-based eye disease diagnosis using visual aids focusing on deep learning methods and smart devices for the visually impaired [8]. The study of low-cost assistive system using deep learning for real-time object detection and environmental description shows enhanced mobility for the visually impaired person [2]. Several studies focus on infrared navigation for blind individuals and mobile robots, investigating obstacle detection, particularly corners, using theoretical, numerical, and experimental methods [9]. A speech recognition system matches the spoken name to the meal database which demonstrated a voice-controlled insulin bolus dose calculator, though untested on visually impaired diabetics [10]. A study presents a footpath dataset of Dhaka, fine-tunes state-of-the-art object detection models, and develops a heuristic navigation algorithm for real-time, speech-based guidance [11]. It has been noted that systems incorporating computer vision for obstacle detection and GPS for navigation often fail to account for the dynamic nature of obstacles and lack real-time path guidance in densely populated environments [1]. The assistive technologies encompass a broad range of applications, such as mobility, navigation, object and activity recognition, information access, interaction, education, and wearable or handheld devices [12].

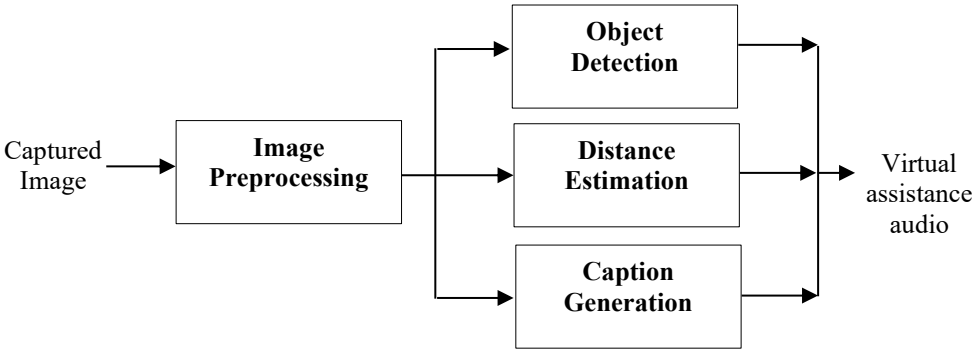

Fig.1. Proposed Visual Assistance System

3. Methods

The Virtual Assistance system consists of three functions, which are caption generation of the view in front of the user, generating the user overall visual information, depth estimation of object identifies the distance of objects from the user and object detection, which specifies the type of object in view of user. The Virtual Assistance System is shown in Fig. 1.

3.1 Finding

Object localization locates boundaries, whereas image classification finds objects. Real-time object recognition and naming is achieved with YOLO and R-CNNs.

3.2 Depth Guess

In 2D images, depth estimation calculates the distances between objects. By using monocular or stereo approaches along with supervised learning to lessen regression loss from monocular pictures, it illustrates 3D structure and spatial layout.

3.3 Captions

CNN served as the model for retrieving visual data, while an LSTM generated the text for the picture captioning feature of the system. Using visual cues from images that were linked to relevant terms from the training dataset—image and caption pairs—the CNN-LSTM model was trained. The model is expressed as

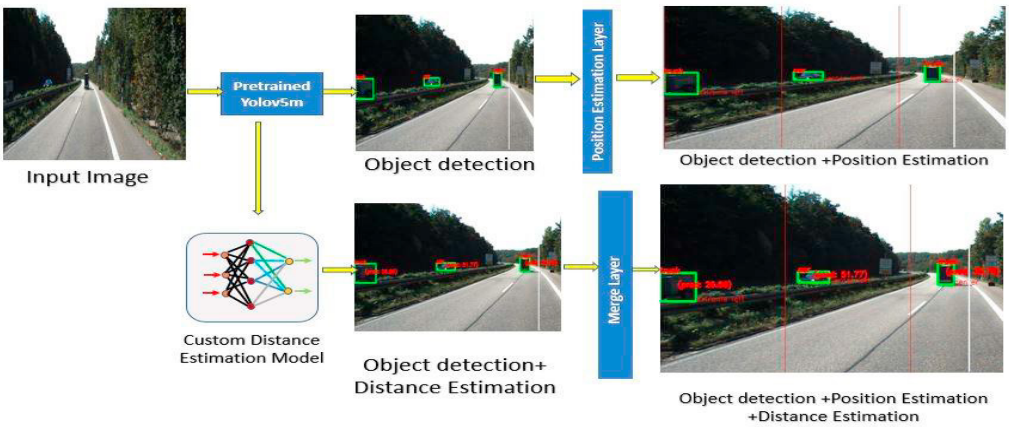

Fig. 2. Function of each component in the proposed system

Table 1. Dataset statistics

Task	Caption Generation	Object Distance Estimation
Dataset	Flicker8K	KITTI
# Train Images	8000	26,000

Image Caption Generator Model = CNN + LSTM

The architecture of the system allows it to generate natural language descriptions for new images.

4. Proposed Architecture

The function of each component in the proposed system is shown in Fig. 2.

4.1 Dataset

This paper uses Flicker8K to train image caption generator models. Eight thousand images and forty thousand captions. Five captions clarify each image. Use the KITTI dataset to estimate object distance and location. Stereo images and 3D laser scans of outside scenery from vehicle-mounted equipment are in this dataset [13]. At 1241 × 376 resolution, the RGB photographs have 26K images. Dataset statistics are shown in Table 1.

4.2 Proposed Model

An internet-based virtual guide system for vision-impaired people sends the user's phone or wearable camera image to the server, where three modules extract captions, distances, and names of outdoor items. The server will send user device text details, which it will convert to audio. The Yolov5 object identification model is pre-trained to identify images and provide bounding boxes for all items on the server. Bounding box data are an object's Xmin, Xmax, Ymin, and Ymax. Customized distance estimation neural networks calculate object distances from object bounding box data using box coordinates. Detailed sequence of subtasks in the proposed architecture is shown in Fig. 3.

4.3 Object Detection and Distance Estimation

Images are confidently labelled and classified using pre-trained models. Using YOLOv5 bounding box data, the position estimation layer uses object position to label five vertical sections. Object distance (z-loc) is measured using a custom model trained on the KITTI dataset. The output includes position, relative distance, and object classification.

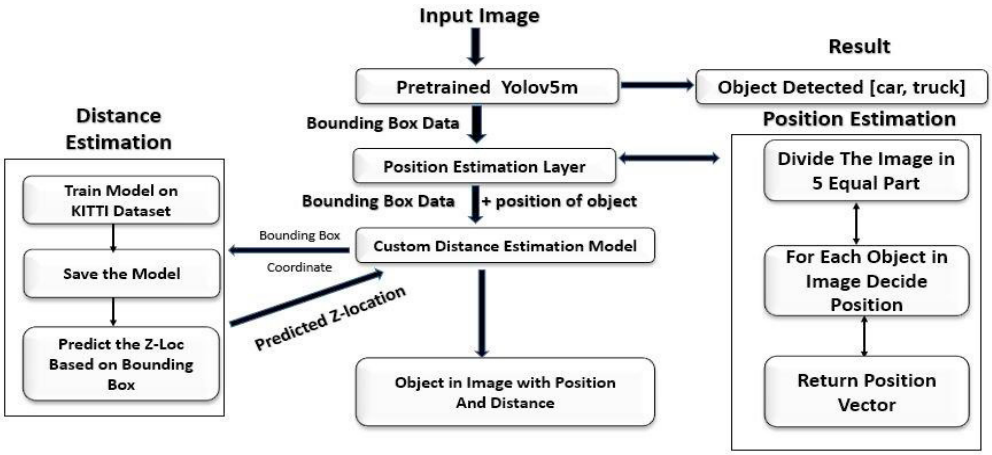

Fig. 3. Detailed sequence of subtasks in the proposed architecture

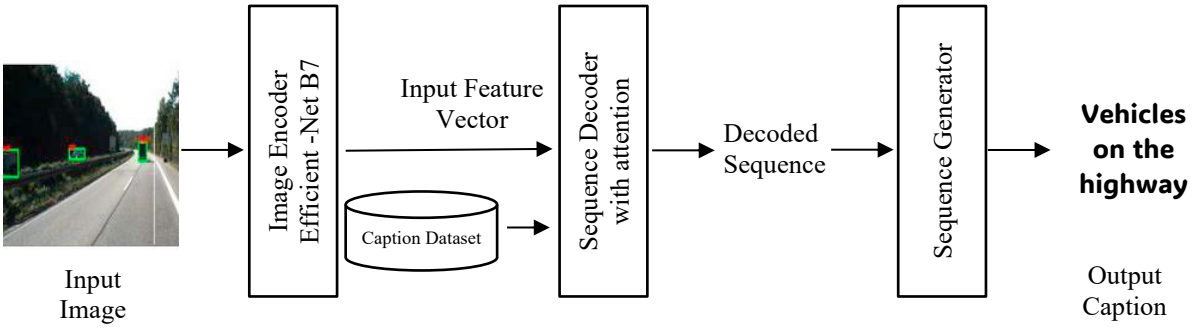

Fig. 4. Caption generation with attention mechanism

4.4 Position Estimation

When an object is positioned, it is positioned at an extreme Left, Left of Center, Center, Right of Center, or Right. YOLOv5 bounding box coordinates are used to partition the image into five parts, and objects are located by comparing the boundaries.

4.5 Caption Generation with Attention Mechanism

Using EfficientNet-B7 to extract features and convert an image to text is the process of captioning. Getting the necessary data, extracting features, creating an encoder-decoder model that incorporates attention, training the model, and then creating captions are all steps in the process. But instead of providing precise navigational advice, our approach describes the scene in a more broadly. This kind of approach typically provides a broad framework and room for improvement. This is a foundational model for generic scope and can be scaled for different scenarios by regress training and testing. Caption generation with attention mechanism is shown in Fig. 4.

Algorithm 1

```
Input: Capture Image, Output: Caption audio
function ProcessImage(I)
Send I to server
O ← YOLOv5(I) // Object detection
for each object o in O do
d ← DistanceEstimation(o.boundingBox) // Custom neural network
p ← DeterminePosition(o.boundingBox) // Left, Center, Right
o.metadata ← {d, p}
end for
C ← CaptionGeneration(I) // EfficientNet-B7 + LSTM with
attention
return {O, C}
end function
function GenerateAudioFeedback(O, C)
F ← CompileInformation(O, C)
A ← TextToSpeech(F)
return A
end function
I ← CaptureImage()
{O, C} ← ProcessImage(I)
A ← GenerateAudioFeedback(O, C)
PlayAudio(A)
```

The Algorithm 1 gives the complete procedure followed in Virtual Assistance System.

5. Results

5.1 Evaluation Metrics

The term Mean Squared Error (MSE) refers to square of the difference between real and predicted values. Mean squared error is used to determine how well predictions or estimates match actual values. The lower the Mean squared error, the more accurate the forecast is.

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \underline{Y}_i)^2 \tag{1}$$

The ratio between the real and expected values can be measured via the mean squared logarithmic error (MSLE). As the name implies, the mean squared logarithmic error is a variant of the mean squared error.

$$L(y,\hat{y}) = \frac{1}{N}\sum_{i=0}^{N}(log\ (y_i + 1) - log\ (\widehat{y_i} + 1))^2 \tag{2}$$

One of the methods most frequently used to assess the accuracy of forecast is the root mean square error, also known as root mean square deviation.

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{N}|y_i - \widehat{y_i}||^2}{N}} \tag{3}$$

BLEU implies for Bilingual Evaluation Understudy, this is the benchmark for evaluating translation of the text with reference translation. It can be applied to examine text generated using natural language processing.

5.2 Object Detection and Distance Estimation Results

For the object detection task Yolov5 object detection model is used which is successor of Yolov4. Yolov5 is 88% smaller and 180% faster than YOLOv4. When we pass an input image to YOLOv5 we get feature vector containing 7 features, Xmin, Ymin, Xmax, Ymax, confidence of the object detected, class of the object and name of the object. Table 2 shows the comparative on the KITTI dataset.

This feature vector is provided as input to custom distance estimation model to provide the depth information of object in image. Depth of the object is estimated based on Xmin, Ymin, Xmax and Ymax.

Table 2. Performances comparison on the KITTI dataset

Model	Parameter	Sq Rel	RMS	RMS Log
Gan et al [14]	-	0.666	3.93	0.173
Enhanced Classifier [15]	40	1.844	6.87	0.314
Ali et al [16]	-	2.020	6.24	0.30
Bhat et al [4]	78	0.190	2.36	0.088
M. Welponer [17]	58	-	5.77	.0330
Proposed Model	**2.96**	**0.163**	3.87	**.0232**

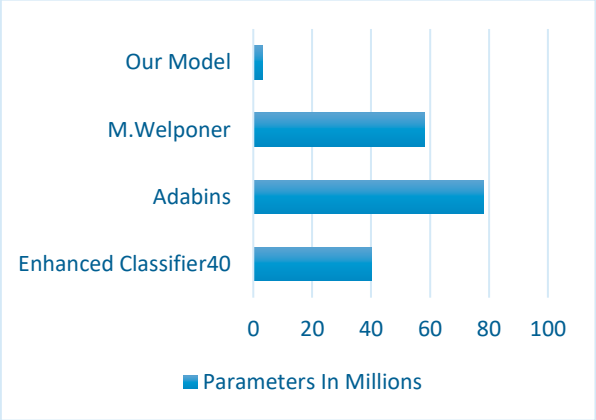

Fig. 5. Comparative analysis of trainable parameters of models over Kitty dataset

The bespoke deep learning model, including 0.296 million parameters and trained on a dataset of 36,000 samples, utilizes a six-layer network and fully connected convolutional layers to make estimations of distances. It has a reduction of 195% in parameters compared to M. Welponer [17], a reduction of 263% compared to AdaBins (Bhat et al) [4], and a reduction of 135% compared to Enhanced Classifier [15]. Comparative research reveals that it exhibits much lower relative error and higher accuracy on the Kitty dataset compared to earlier models. Comparative Analysis of No of Trainable parameter of Models over Kitty Dataset is shown in Fig. 5. It can be noted that the proposed model outperformed with a large margin in all of the 4 parameters, while it outperformed Adabins in 3 out of 4 parameters, where its root mean squared of error is 3.87 while Adabins has 2.36. Graphical representation for comparative analysis of model performance over Kitty dataset is shown in Fig. 6.

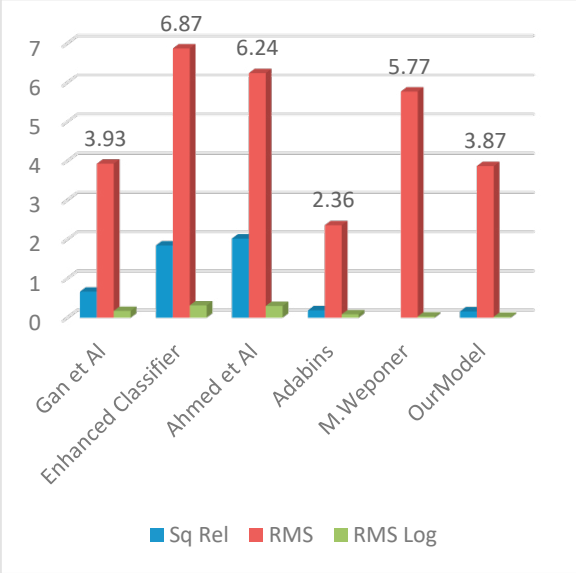

Fig. 6. Comparative performance analysis of models over Kitty dataset

Table 3. Comparison of performances on the Flickr8k dataset

Model	BLEU-1	BLEU-2	BLEU-3	BLEU-4
Balachandar, A., et al [18]	0.56	-	-	-
Zhang et al [19]	0.61	0.45	0.36	0.26
M. Welponer [17]	0.63	0.40	0.28	0.15
Proposed Model	**0.65**	**0.57**	**0.48**	**0.39**

C. Caption Generation Results

The proposed caption generation model is trained on flickr8k dataset, and it generate the BLEU 1,2,3,4 score 0.6474, 0.57, 0.4832, 0.3853 respectively. The proposed system can generate better results in comparison with other reference paper in performance analysis. Comparative performance on Flickr8k dataset is shown in Table 3 as well as its graphical representation is shown in Fig. 7.

6. Conclusion and Future work

This work introduces a new type of visual aid system designed specifically to be appropriate for visually impaired people in outdoor environments. It includes YOLOv5 for effective object detection, a custom model for distance and position estimation that outperforms previous models, and an attention mechanism with caption generation. Rather than providing clear navigational instructions, it now provides greater and broader definitions of the surroundings, which can be the foundation for worldwide contextual visual aid systems. The sights are presented in generic terms to better explain the surroundings, as opposed to the specifics of "Turn left for Delhi," which is lacking. Future work will build on this foundation by adding real-time audio feedback, live video feed input, and possibly even moving toward more specialized navigation assistance through the use of sophisticated natural language processing techniques, GPS and satellite data, navigation-oriented datasets, and so forth. The system's adaptation for indoor surroundings is a third potential avenue for future research, since it could enable visually impaired individuals to utilize it in a wider range of contexts. This is significant as a first step toward more extensive and context-aware help systems; overall scene awareness is weighed against the potential for more, more focused navigational support in subsequent versions.

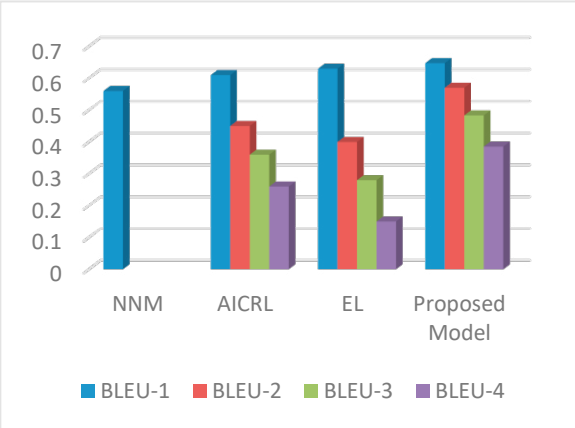

Fig. 7. Comparative analysis of BLEU scores of models

References

1. Scalvini, F., Bordeau, C., Ambard, M., Migniot, C., & Dubois, J. (2023). Outdoor navigation assistive system based on robust and real-time visual–auditory substitution approach. *Sensors*, *24*(1), 166.
2. Islam, R. B., Akhter, S., Iqbal, F., Rahman, M. S. U., & Khan, R. (2023). Deep learning based object detection and surrounding environment description for visually impaired people. *Heliyon*, *9*(6).
3. Bauer, Z., Dominguez, A., Cruz, E., GomezSq F., Orts-Escolano, S., & Cazorla, M. (2020). Enhancing perception for the visually impaired with deep learning techniques and lowcost wearable sensors. Pattern recognition letters, 137, 27-36.
4. Bhat, S. F., Alhashim, I., & Wonka, P. (2021). Adabins: Depth estimation using adaptive bins. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 4009-4018).
5. Bahdanau, D., Cho, K., & Bengio, Y. (2014). Neural machine translation by jointly learning to align and translate. arXiv preprint arXiv:1409.0473.
6. Patil, K., Kharat, A., Chaudhary, P., Badgar, S., & Givhan, R. (2021, March). Guidance System for Visually Impaired People. In 2021 International Conference on Artificial Intelligence and Smart Systems (ICAIS) (pp. 988-993). IEEE.
7. Hsieh, I., Cheng, H. C., Ke, H. H., Chen, H. C., & Wang, W. J. (2021). A CNN-Based Wearable Assistive System for Visually Impaired People Walking Outdoors. Applied Sciences, 11(21), 10026.
8. Wang, J., Wang, S., & Zhang, Y. (2023). Artificial intelligence for visually impaired. *Displays*, *77*, 102391.
9. Papagianopoulos, I., De Mey, G., Kos, A., Wiecek, B., & Chatziathasiou, V. (2023). Obstacle detection in infrared navigation for blind people and mobile robots. *Sensors*, *23*(16), 7198.
10. Muhsin, Z. J., Qahwaji, R., Ghanchi, F., & Al-Taee, M. (2024). Review of substitutive assistive tools and technologies for people with visual impairments: recent advancements and prospects. *Journal on Multimodal User Interfaces*, *18*(1), 135-156.
11. Chowdhury, R. I., Anjom, J., & Hossain, M. I. A. (2024). A novel edge intelligence-based solution for safer footpath navigation of visually impaired using computer vision. *Journal of King Saud University-Computer and Information Sciences*, 102191.
12. Ketema Dabi, G., & Negassa Golga, D. (2024). The role of assistive technology in supporting the engagement of students with visual impairment in learning mathematics: An integrative literature review. *British Journal of Visual Impairment*, *42*(3), 674-687.
13. Lin, Y., Wang, K., Yi, W., & Lian, S. (2019). Deep learning based wearable assistive system for visually impaired people. In Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops.
14. Gan, Y., Xu, X., Sun, W., & Lin, L. (2018). Monocular depth estimation with affinity, vertical pooling, and label enhancement. In *Proceedings of the European Conference on Computer Vision (ECCV)* (pp. 224-239).
15. Jamtsho, Y., Riyamongkol, P., & Waranusast, R. . (2019). Real-time bhutanese license plate localization using yolo. ICT Express,6(2)
16. Ali, A., Hassan, A., Ali, A. R., Khan, H. U., Kazmi, W., & Zaheer, A. (2020). Real-time vehicle distance estimation using single view geometry. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision* (pp. 1111-1120).
17. Welponer, M., Stathopoulou, E. K., & Remondino, F. (2022). MONOCULAR DEPTH PREDICTION IN PHOTOGRAMMETRIC APPLICATIONS. *The International Archives of Photogrammetry, Remote Sensing and Spatial Information Sciences*, *43*, 469-476.
18. Balachandar, A., Santhosh, E., Suriyakrishnan, A., Vigensh, N., Usharani, S., & Bala, P. M. (2021, May). Deep Learning Technique Based Visually Impaired People Using YOLO V3 Framework Mechanism. In 2021 3rd International Conference on Signal Processing and Communication (ICPSC) (pp. 134-138). IEEE.
19. Zhang, W., Tang, S., Su, J., Xiao, J., & Zhuang, Y. (2021). Tell and guess: cooperative learning for natural image caption generation with hierarchical refined attention. Multimedia Tools and Applications, 80(11), 16267-16282.